

Challenge 2: Excluded Minors of $\text{GF}(p^r)$ -representable matroids

Definition 1 (Matroid). A *matroid* is a pair $M = (E, \mathcal{I})$ consisting of a finite set E and a collection $\mathcal{I}$ of subsets of E , called the *independent sets*, such that:

1. $\emptyset \in \mathcal{I}$;
2. if $I \in \mathcal{I}$ and $J \subseteq I$, then $J \in \mathcal{I}$;
3. if $I, J \in \mathcal{I}$ and $|I| < |J|$, then there is some $x \in J \setminus I$ such that $I \cup \{x\} \in \mathcal{I}$.

The maximal independent sets of M are called the *bases* of M .

Definition 2 ($\text{GF}(p^r)$ -representable matroid). Let p be a prime number. Let r be a positive integer. Let $\text{GF}(p^r)$ denote the finite field with p^r elements. A matroid M is $\text{GF}(p^r)$ -*representable* if there is a matrix A over $\text{GF}(p^r)$ such that the independent sets of M are exactly the sets of columns of A which are linearly independent over $\text{GF}(p^r)$.

Definition 3 (Matroid minor). Let M be a matroid with ground set E . The *dual* of M , denoted M^* , is the matroid with the same ground set, where the bases are the complements of the bases of M . Given a subset $D \subseteq E$, the *deletion* of D from M , denoted $M \setminus D$, is the restriction of M to $E \setminus D$. Given a subset $C \subseteq E$, the *contraction* of M onto $E - C$ is the matroid $M/C := (M^* \setminus C)^*$, with ground set $E \setminus C$.

Given matroids M, N , then N is a *minor* of M if there exist sets C and D such that $N = M/C \setminus D$.

Definition 4 (Excluded minor). Let $\mathcal{C}$ be a class of matroids closed under minors. A matroid M is an *excluded minor* for $\mathcal{C}$ if $M \notin \mathcal{C}$, but every proper minor of M belongs to $\mathcal{C}$.

The class of $\text{GF}(p^r)$ -representable matroids is closed under taking minors. Thus, it is natural to ask for minimal matroids which fail to be representable over $\text{GF}(p^r)$.

Open Problem 1.1. Let p be a prime number. Let r be a positive integer. List the minimal excluded minors for the class of $\text{GF}(p^r)$ -representable matroids and prove that the given list is complete up to isomorphism.

For $p^r = 2$, the excluded minor list is known: a matroid is representable over $\text{GF}(2)$ if and only if it has no minor isomorphic to $U_{2,4}$ [1]. For $p^r = 3$, there is also a finite explicit list: a matroid is representable over $\text{GF}(3)$ if and only if it has no minor isomorphic to $U_{2,5}$, $U_{3,5}$, F_7 , or F_7^* [2, 3]. In general, Rota's excluded-minor conjecture, proven by Geelen, Gerards and Whittle, says that for every finite field $\text{GF}(p^r)$ there are only finitely many excluded minors for the class of $\text{GF}(p^r)$ -representable matroids [4, 5].

We have the following parametrized version of the open problem.

Open Problem 1.2. Let r be a positive integer. If there is a prime number p and a positive integer m such that $r = p^m$, then list the minimal excluded minors for the class of $\text{GF}(p^m)$ -representable matroids and prove that the list is complete up to isomorphism.

In the formal statement, the list is given as a finite set L of matroids: the members of L are pairwise non-isomorphic excluded minors, and every finite matroid that is not $\text{GF}(p^m)$ -representable has a minor isomorphic to a member of L . This rules out indirect solutions that prove the existence of a complete list without exhibiting one.

References

- [1] W. T. Tutte. “A homotopy theorem for matroids. I, II”. In: *Transactions of the American Mathematical Society* 88 (1958), pp. 144–174.
- [2] R. E. Bixby. “On Reid’s characterization of the ternary matroids”. In: *Journal of Combinatorial Theory, Series B* 26.2 (1979), pp. 174–204.
- [3] P. D. Seymour. “Matroid representation over $\text{GF}(3)$ ”. In: *Journal of Combinatorial Theory, Series B* 26.2 (1979), pp. 159–173.
- [4] G.-C. Rota. “Combinatorial theory, old and new”. In: *Actes du Congrès International des Mathématiciens, Nice, 1970*. Vol. 3. Paris: Gauthier-Villars, 1971, pp. 229–233.
- [5] J. Geelen, B. Gerards, and G. Whittle. “Solving Rota’s conjecture”. In: *Notices of the American Mathematical Society* 61.7 (2014), pp. 736–743. DOI: 10.1090/noti1139.